requires the determination of electronic charge distribution within the atom. It may, however, be noted that in the calculations involving quantum mechanics, the concept of orbits is replaced by the wave function and the quantity $\langle r^2 \rangle$ has a different significance. The experimental results agree fairly well with the theoretically calculated values particularly for light elements and rare gases, whereas large deviations exist for ions containing a very large number of electrons. The typical value for helium is about -1.9×10^{-6} which shows that the diamagnetism is a very weak effect.

8.4 PARAMAGNETISM

Paramagnetism occurs in those atoms, ions and molecules which have permanent magnetic moments. In the absence of a magnetic field, these magnetic moments are oriented randomly and no net magnetization is produced. When a magnetic field is applied, these moments orient themselves in the direction of the field which results in some net magnetization parallel to the applied field. The paramagnetic materials have small, positive and temperature-dependent susceptibility. The permanent magnetic moments of ions result from the following contributions :

(i) The spin or intrinsic moments of the electrons,

(ii) The orbital motion of the electrons,

(iii) The spin magnetic moment of the nucleus.

Paramagnetism is observed in :

(i) Metals

(ii) Atoms and molecules possessing an odd number of electrons, e.g., free sodium atoms, gaseous nitric oxide, etc.

(iii) A few compounds having an even number of electrons, e.g., oxygen molecule.

(iv) Free atoms and ions having a partially filled inner shell, e.g., rare earth and actinide elements, ions of some transition elements such as Mn^{2+}, etc.

8.4.1 Langevin's Classical Theory

Langevin considered a paramagnetic gas containing N atoms per unit volume each having a permanent magnetic moment μ. The mutual interaction between the magnetic dipoles was as-

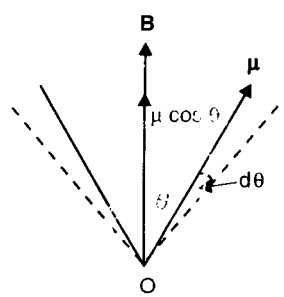


Fig. 8.1 *Magnetic dipole with moment* μ *oriented at an angle* θ *to the applied magnetic field.*

sumed to be negligible. In the presence of a magnetic induction **B**, these dipoles tend to orient themselves in the direction of the field in order to minimize their energy. The thermal energy at ordinary temperatures, however, resists such an alignment of dipoles. In thermal equilibrium, the dipoles orient themselves at an angle θ with the direction of the applied field as shown in Fig. 8.1. The potential energy of each dipole in this position is given by

$$E = -\mu.\mathbf{B} = -\mu B \cos\theta$$

Using Maxwell-Boltzmann distribution law, the number of magnetic dipoles having this particular orientation is proportinal to

$$\exp\left(-\frac{E}{kT}\right) \quad \text{or} \quad \exp\left(\frac{\mu B\cos\theta}{kT}\right)$$

Also, according to statistical mechanics, the probability for a magnetic dipole to make an angle between θ and $\theta + d\theta$ with the magnetic field, or the number of dipoles, dn, having axes within the solid angle $d\omega$ lying between two hollow cones of semiangles θ and $\theta + d\theta$ (Fig. 8.1) is given by

$$dn \propto \exp\left(\frac{\mu B\cos\theta}{kT}\right) d\omega$$

$$= k \exp\left(\frac{\mu B\cos\theta}{kT}\right) 2\pi \sin\theta\, d\theta \tag{8.29}$$

where k is a constant. Each one of these dipoles contributes a component of magnetic moment $\mu \cos\theta$ to the magnetization, whereas the components perpendicular to the field direction cancel each other. Hence the average component of the magnetic moment of each atom along the field direction multiplied by the number of atoms per unit volume, N, gives the magnetization M, i.e.,

$$M = N\mu\langle\cos\theta\rangle = N\frac{\int_0^\pi \mu\cos\theta\, dn}{\int_0^\pi dn} = N\frac{\mu\int_0^\pi \cos\theta\sin\theta\exp\left(\frac{\mu B\cos\theta}{kT}\right)d\theta}{\int_0^\pi \sin\theta\exp\left(\frac{\mu B\cos\theta}{kT}\right)d\theta} \tag{8.30}$$

Let $\quad \frac{\mu B}{kT} = x,$

and $\quad \cos\theta = y$

$\therefore \quad -\sin\theta\, d\theta = dy$

Therefore, Eq. (8.30) becomes

$$M = N\mu \frac{\int_1^{-1} y e^{xy} dy}{\int_1^{-1} e^{xy} dy} = N\mu \left[\frac{e^x + e^{-x}}{e^x - e^{-x}} - \frac{1}{x} \right] = N\mu \left[\coth x - \frac{1}{x} \right]$$

$$= N\mu\, L(x) \tag{8.31}$$

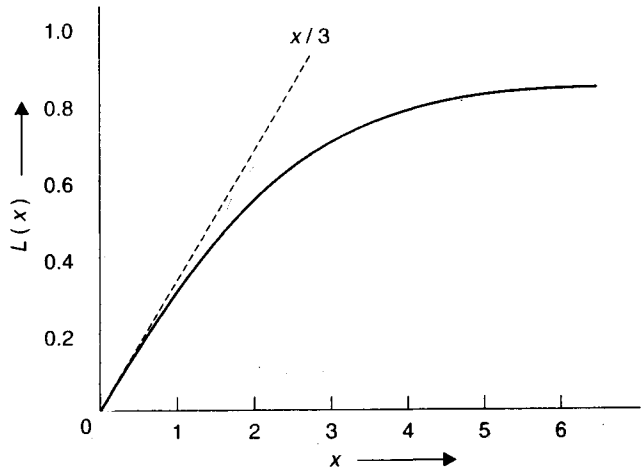


Fig. 8.2. *Variation of L(x) with x.*

where $L(x)$ is called the *Langevin function.* The variation of $L(x)$ with $x = \dfrac{\mu B}{kT}$ is shown in Fig. 8.2. For $x << 1$, i.e., at normal field strengths and ordinary temperatures, the curve is almost linear and coincides with the tangent to the curve at the origin which is equal to $x/3$. Thus, we have

$$L(x)\big|_{x=0} \cong \frac{x}{3} = \frac{\mu B}{3kT} \tag{8.32}$$

Therefore, Eq. (8.31) gives

$$M \cong \frac{N\mu^2 B}{3kT}$$

The *paramagnetic susceptibility* is given as

$$\chi_{para} = \frac{\mu_0 M}{B} = \frac{\mu_0 N\mu^2}{3kT} \tag{8.33}$$

which shows that the susceptibility is inversely proportional to temperature. It can be written as

$$\chi_{para} = \frac{C}{T} \tag{8.34}$$

where $C = \dfrac{\mu_0 N \mu^2}{3k}$ is a constant. The expression (8.34) is called the *Curie law* and C is called the *Curie constant.* Thus we find that the Curie law holds good for $\mu B << kT$.

For large values of x, i.e., for high flux densities and low temperatures, $L(x)$ approaches unity and Eq. (8.31) gives

$$M = N\mu = M_s \tag{8.35}$$

This is the saturation condition which corresponds to the complete alignment of the magnetic dipoles in the field direction; M_s is called the *saturation magnetization.*

8.4.2 Quantum Theory

The classical theory assumes that the permanent magnetic moment of a given atom or ion rotates freely and can possess any orientation with respect to the applied magnetic field. According to quantum theory, since these magnetic moments are quantized, the magnetic dipole moment $\boldsymbol{\mu}$ and its component μ_z in the direction of the applied field cannot have arbitrary values. We have, in general, a direct relationship between the magnetic dipole moment $\boldsymbol{\mu}$ of an atom or ion in free space and its angular momentum $\mathbf{J}$ as

$$\boldsymbol{\mu} = -g\mu_B \mathbf{J} \tag{8.36}$$

The quantity μ_B is called the Bohr magneton and is equal to $\dfrac{e\hbar}{2m}$ in SI system and $\dfrac{e\hbar}{2mc}$ in CGS system of units; g is known as the Lande's g-factor and is equal to 2 if the net angular momentum of the dipole is due to electron spin and 1 if it is due to orbital motion only. In general, it has mixed origin and is obtained from the expression :

$$g = 1 + \frac{J(J+1) + S(S+1) - L(L+1)}{2J(J+1)} \tag{8.37}$$

where S and L represent the spin and orbital quantum numbers of the dipole respectively. The orientations of the magnetic moment $\boldsymbol{\mu}$ with respect to the direction of the applied magnetic field are specified by the rule that the possible components of $\boldsymbol{\mu}$ along the field direction are given by

$$\mu_z = -g\mu_B m_J \tag{8.38}$$

where $m_J = -J, -J+1, \ldots, J-1, J$ is the magnetic quantum number associated

with J. For each value of J, m_J can have $(2J+1)$ values which means that the magnetic moment of the atom can have $(2J+1)$ different orientations relative to the field. The potential energy of such a magnetic dipole in the presence of a magnetic field B is, therefore, given by

$$E = m_J g \mu_B B \tag{8.39}$$

According to the Maxwell-Boltzmann distribution, the number of atoms having a particular value of m_J is thus proportional to

$$e^{-m_J g \mu_B B/kT}$$

Considering a unit volume of a paramagnetic material containing a total of N atoms, the magnetization in the direction of the field is given by

$M = N \times$ (Statistical average of the magnetic moment component per atom along B)

$$= N \sum_{m_J=-J}^{m_J=+J} \frac{-m_J g \mu_B e^{-m_J g \mu_B B/kT}}{e^{-m_J g \mu_B B/kT}} \tag{8.40}$$

We consider the following two cases :

(i) At normal flux densities and ordinary temperatures,

$$\frac{m_J g \mu_B B}{kT} << 1$$

Therefore, the expression (8.40) can be approximated as

$$M = N \frac{g\mu_B \sum_{m_J=-J}^{+J} -m_J \left(1 - \frac{m_J g \mu_B B}{kT}\right)}{\sum_{m_J=-J}^{+J} \left(1 - \frac{m_J g \mu_B B}{kT}\right)}$$

Now, $$\sum_{m_J=-J}^{+J} m_J = 0$$

and $$\sum_{m_J=-J}^{+J} m_J^2 = 2 \sum_{m_J=0}^{J} m_J^2 = \frac{J(J+1)(2J+1)}{3}$$

$\therefore$ $$M = N \frac{\frac{g^2 \mu_B^2 B}{kT} \cdot \frac{J(J+1)(2J+1)}{3}}{(2J+1)}$$

$$= N \frac{g^2 \mu_B^2 B}{3kT} J(J+1) \tag{8.41}$$

$$\therefore \quad \chi_{para} = \frac{\mu_o M}{B} = \frac{\mu_o N \mu_B^2}{3kT} g^2 J(J+1) \tag{8.42}$$

or
$$\chi_{para} = \frac{\mu_o N p_{eff}^2 \mu_B^2}{3kT} \tag{8.43}$$

where p_{eff} is the effective number of Bohr magnetons and is given by

$$p_{eff} = g\sqrt{J(J+1)} \tag{8.44}$$

The expression (8.43) is identical to the classical expression (8.33) with

$$\mu^2 = p_{eff}^2 \mu_B^2 \tag{8.45}$$

The Curie law can also be deduced from Eq. (8.43) as described earlier.

(ii) At low temperatures and strong magnetic fields, $\frac{m_J g \mu_B B}{kT}$ is not smaller than unity and it is not possible to make a series expansion of the exponential terms in the expression (8.40). After some algebraic manipulations, Eq. (8.40) yields

$$M = NgJ\mu_B B_J(x) \tag{8.46}$$

where $x = \frac{gJ\mu_B B}{kT}$ and $B_J(x)$ is the Brillouin function defined as

$$B_J(x) = \frac{2J+1}{2J} \coth\left(\frac{2J+1}{2J}\right)x - \frac{1}{2J}\coth\left(\frac{x}{2J}\right) \tag{8.47}$$

For $x << 1$, we have

$$\coth x \cong \frac{1}{x} + \frac{x}{3}$$

$$\therefore \quad B_J(x) = \frac{x(J+1)}{3J}$$

Thus the susceptibility becomes

$$\chi_{para} = \frac{\mu_o M}{B} = \frac{\mu_o N J(J+1) g^2 \mu_B^2}{3kT} = \frac{\mu_o N p_{eff}^2 \mu_B^2}{3kT} \tag{8.48}$$

which is the same as (8.43). For $x >> 1$. i.e., at low temperatures and strong fields,

$$\coth x \cong 1$$

$$\therefore \quad B_J(x) \cong 1$$

and Eq. (8.46) becomes

$$M = NgJ\mu_B \tag{8.49}$$

This result implies the state of magnetic saturation, i.e., all the dipoles get aligned along the magnetic induction **B**. Thus, in this limit, the expression (8.46) is analogous to the Langevin expression (8.31) with the difference that the latter is applicable to freely rotating dipoles only. In fact, for $J \to \infty$, i.e., for a large number of allowed orientations of a magnetic dipole, we have

$$\coth \frac{x}{2J} \to \frac{2J}{x}$$

and

$$\coth \left(1+\frac{1}{2J}\right)x \to \coth x$$

$$\therefore \quad B_J(x) \to \coth x - \frac{1}{x} \text{ or } L(x)$$

Thus the quantum results approach the classical ones. This is what is expected as the classical theory allows all conceivable orientations. For other values of J, however, the two results differ considerably.

The order of magnitude of paramagnetic susceptibility of a solid as estimated from Eq. (8.48) at room temperature is about 10^{-7} which is quite small. It increases by more than hundred times at 1 K. Equation (8.48) is successfully employed to predict the values of susceptibility for various paramagnetic crystals particularly rare earth ions. The value of J is determined by applying the Hund's rules [1]. However, Eq. (8.48) is unable to account for the experimental observations of susceptibility for the ions of the iron group. This is because of the presence of crystal field due to other ions which

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(1) Hund's rules state that, for the ground state of atoms with incompletely filled shells,

(i) the electron spins add up to give the maximum possible S consistent with the Pauli's exclusion principle.

(ii) the orbital momenta combine to give the maximum value for L that is consistent with (i)

(iii) the value of J is given by
$J = |L - S|$ if a shell is less than half-filled
$J = L + S$ if a shell is more than half-filled.
If the shell is just half-filled, $L = 0$ and, therefore, $J = S$